```
!pip install -q spacy
```

```
!python -m spacy download en_core_web_sm
```

```
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_cor
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 75.2 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import spacy
from spacy import displacy
```

```
nlp = spacy.load("en_core_web_sm")

print("spaCy model loaded successfully!")
```

```
spaCy model loaded successfully!
```

```
text = """
Apple Inc. announced on September 1, 2026 that it will invest $5 billion
in a new technology center in California. CEO Tim Cook said the company
will create more than 10,000 jobs over the next five years.

The new center will be located near San Francisco and will focus on
artificial intelligence and semiconductor research. Microsoft and Google
are also expanding their AI operations in the United States.
"""
```

```
doc = nlp(text)

print("Text processed successfully!")
```

```
Text processed successfully!
```

```
print("========== NAMED ENTITIES ==========\n")

for ent in doc.ents:
    print(
        f"Entity: {ent.text:<25} "
        f"Label: {ent.label_:<10} "
```

```
        f"Description: {spacy.explain(ent.label_)}"
    )
```

```
========== NAMED ENTITIES ==========

Entity: Apple Inc.                Label: ORG        Description: Companies, agen
Entity: September 1, 2026         Label: DATE       Description: Absolute or rel
Entity: $5 billion                Label: MONEY      Description: Monetary values
Entity: California                Label: GPE        Description: Countries, citi
Entity: Tim Cook                  Label: PERSON     Description: People, includi
Entity: more than 10,000          Label: CARDINAL   Description: Numerals that d
Entity: the next five years       Label: DATE       Description: Absolute or rel
Entity: San Francisco             Label: GPE        Description: Countries, citi
Entity: Microsoft                 Label: ORG        Description: Companies, agen
Entity: Google                    Label: ORG        Description: Companies, agen
Entity: AI                        Label: GPE        Description: Countries, citi
Entity: the United States         Label: GPE        Description: Countries, citi
```

```
for ent in doc.ents:
    print(ent.text, "→", ent.label_)
```

```
Apple Inc. → ORG
September 1, 2026 → DATE
$5 billion → MONEY
California → GPE
Tim Cook → PERSON
more than 10,000 → CARDINAL
the next five years → DATE
San Francisco → GPE
Microsoft → ORG
Google → ORG
AI → GPE
the United States → GPE
```

```
print(spacy.explain("ORG"))
print(spacy.explain("GPE"))
print(spacy.explain("PERSON"))
print(spacy.explain("MONEY"))
```

```
Companies, agencies, institutions, etc.
Countries, cities, states
People, including fictional
Monetary values, including unit
```

```
displacy.render(
    doc,
    style="ent",
    jupyter=True
)
```

Apple Inc. **ORG** announced on September 1, 2026 **DATE** that it will invest $5 billion **MONEY** in a new technology center in California **GPE** . CEO Tim Cook **PERSON** said the company will create more than 10,000 **CARDINAL** jobs over the next five years **DATE** .

The new center will be located near San Francisco **GPE** and will focus on artificial intelligence and semiconductor research. Microsoft **ORG** and Google **ORG** are also expanding their AI **GPE** operations in the United States **GPE** .

```
people = [
    ent.text
    for ent in doc.ents
    if ent.label_ == "PERSON"
]

print("People:")
print(people)
```

```
People:
['Tim Cook']
```

```
organizations = [
    ent.text
    for ent in doc.ents
    if ent.label_ == "ORG"
]

print("Organizations:")
print(organizations)
```

```
Organizations:
['Apple Inc.', 'Microsoft', 'Google']
```

```
locations = [
    ent.text
    for ent in doc.ents
    if ent.label_ in ["GPE", "LOC"]
]

print("Locations:")
print(locations)
```

```
Locations:
['California', 'San Francisco', 'AI', 'the United States']
```

```
import pandas as pd

entities = []

for ent in doc.ents:
    entities.append({
        "Entity": ent.text,
        "Label": ent.label_,
        "Description": spacy.explain(ent.label_)
    })

entity_df = pd.DataFrame(entities)

display(entity_df)
```

	Entity	Label	Description
0	Apple Inc.	ORG	Companies, agencies, institutions, etc.
1	September 1, 2026	DATE	Absolute or relative dates or periods
2	$5 billion	MONEY	Monetary values, including unit
3	California	GPE	Countries, cities, states
4	Tim Cook	PERSON	People, including fictional
5	more than 10,000	CARDINAL	Numerals that do not fall under another type
6	the next five years	DATE	Absolute or relative dates or periods
7	San Francisco	GPE	Countries, cities, states
8	Microsoft	ORG	Companies, agencies, institutions, etc.
9	Google	ORG	Companies, agencies, institutions, etc.
10	AI	GPE	Countries, cities, states
11	the United States	GPE	Countries, cities, states

Next steps: Generate code with `entity_df` New interactive sheet

```
TRAIN_DATA = [
    (
```

```
            "The patient was diagnosed with diabetes and prescribed Metformin.",
            {
                "entities": [
                    (30, 38, "DISEASE"),
                    (56, 65, "MEDICATION")
                ]
            }
        ),

        (
            "The doctor prescribed Insulin to treat diabetes.",
            {
                "entities": [
                    (21, 28, "MEDICATION"),
                    (41, 49, "DISEASE")
                ]
            }
        ),

        (
            "Hypertension can be treated using Amlodipine.",
            {
                "entities": [
                    (0, 12, "DISEASE"),
                    (33, 43, "MEDICATION")
                ]
            }
        ),

        (
            "The patient has asthma and uses Salbutamol.",
            {
                "entities": [
                    (18, 24, "DISEASE"),
                    (35, 45, "MEDICATION")
                ]
            }
        ),

        (
            "Doctors are studying treatments for cancer using chemotherapy.",
            {
                "entities": [
                    (38, 44, "DISEASE"),
                    (50, 62, "TREATMENT")
                ]
            }
        )
    ]
```

```
import spacy

custom_nlp = spacy.blank("en")

print("Blank model created.")
```

```
Blank model created.
```

```
ner = custom_nlp.add_pipe("ner")

# Add custom entity labels
ner.add_label("DISEASE")
ner.add_label("MEDICATION")
ner.add_label("TREATMENT")

print("Custom labels added.")
```

```
Custom labels added.
```

```
from spacy.training import Example

optimizer = custom_nlp.begin_training()

for iteration in range(30):

    losses = {}

    for text, annotations in TRAIN_DATA:

        example = Example.from_dict(
            custom_nlp.make_doc(text),
            annotations
        )

        custom_nlp.update(
            [example],
            drop=0.3,
            losses=losses
        )

    if iteration % 5 == 0:
        print(
            f"Iteration {iteration}, "
            f"Loss: {losses}"
        )
```

```
/usr/local/lib/python3.13/dist-packages/spacy/training/iob_utils.py:149: UserWar
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/spacy/training/iob_utils.py:149: UserWar
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/spacy/training/iob_utils.py:149: UserWar
```

```
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/spacy/training/iob_utils.py:149: UserWar
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/spacy/training/iob_utils.py:149: UserWar
  warnings.warn(
Iteration 0, Loss: {'ner': np.float32(25.68656)}
Iteration 5, Loss: {'ner': np.float32(1.9870881)}
Iteration 10, Loss: {'ner': np.float32(0.73894066)}
Iteration 15, Loss: {'ner': np.float32(0.5087675)}
Iteration 20, Loss: {'ner': np.float32(1.1110076e-05)}
Iteration 25, Loss: {'ner': np.float32(2.9213345e-06)}
```

```
test_text = """
The patient has diabetes and hypertension.
The doctor prescribed Metformin and Amlodipine.
The patient is also receiving chemotherapy.
"""

doc_custom = custom_nlp(test_text)

for ent in doc_custom.ents:
    print(
        f"{ent.text} → {ent.label_}"
    )
```

```
displacy.render(
    doc_custom,
    style="ent",
    jupyter=True
)
```

```
/usr/local/lib/python3.13/dist-packages/spacy/displacy/__init__.py:214: UserWarn
  warnings.warn(Warnings.W006)
```

The patient has diabetes and hypertension.

The doctor prescribed Metformin and Amlodipine.

The patient is also receiving chemotherapy.

```
nlp = spacy.load("en_core_web_sm")

ner = nlp.get_pipe("ner")

ner.add_label("DISEASE")
ner.add_label("MEDICATION")
ner.add_label("TREATMENT")
```

```
1
```

```
text = """
Russian President Vladimir Putin has been urged by Indian Prime Minister Narenc

"Every day spent in war pushes humanity back, and every step taken towards peac

Modi, whose country's economy depends heavily on discounted oil from Moscow, ha

BBC correspondents in Delhi and Moscow look at what lies behind Modi's renewed

Vikas Pandey
India digital editor, Delhi
Modi's comments on ending the war in Ukraine are not surprising and don't sugge

In fact, speaking on the sidelines of a regional summit in 2022 - less than a y

But it would be an oversimplification to suggest that Modi's current or past st

Western countries have pushed Modi hard to take a clear stand on the war - but

The stand hasn't made many Western capitals happy but they have continued to cc

In this context, Modi's current statement cannot be seen as an attempt to pleas

Getty Images Storage tanks at a Bharat Petroleum Corp. oil refinery in Mumbai,
India's economy relies heavily on imported Russian oil

If anything, it is very much in line with India's policy of strategic autonomy.

The answer lies in the long-standing Russia-India ties - the two countries have

Moscow has continued to be one of India's strongest allies despite Delhi forgir

Delhi is one of the founding members of the Brics grouping of nations along wit

But at the same time, Delhi is also a member of the Quad group with Australia,

Still, Moscow and Delhi tend to work through these differences. Moreover, Russi

So, Modi's latest statement is not expected to cause any rift in Moscow-Delhi r


As for the West and Modi's critics in India, they will feel that the prime mini

Steve Rosenberg
Russia editor, Moscow
This isn't the first time a foreign leader has publicly urged Putin to end the

Just five weeks ago the President of Kazakhstan Kassym-Jomart Tokayev suggestec
```

```
He didn't. And he hasn't.

There's no sign of these public calls for peace changing Putin's calculations.

To Modi's call for a cessation of hostilities, the Kremlin leader replied:

"Thank you very much, prime minister. We are moving forward along this path."

Polite, but vague. And non-committal.

After all, a path can be short. But it can also be long. Very long.

And "moving forward" at what pace?

And, crucially, on which path?

All the evidence suggests that Russia's leadership remains convinced it is on a


Therefore, for now, the fighting goes on.

Despite the fact that, after more than four-and-a-half years of war and huge mi

And despite Ukraine's success in bringing the war back home to Russians: for ex

Reuters A woman in Ukraine's town of Boryspil, outside Kyiv, collects belonging
Four people were killed in overnight Russian strikes on Ukraine's town of Borys
In recent days Russia has stepped up its attacks on Ukraine.

"After [Russia] intensified its strikes on Ukraine's military-industrial facili

While Putin believes that, he'll be reluctant to pause his so-called "special m
"""
```

```
doc = nlp(text)

for ent in doc.ents:
    print(ent.text, "→", ent.label_)
```

```
Moscow → GPE
Putin → PERSON
West → LOC
Modi → GPE
India → GPE
Steve Rosenberg → PERSON
Russia → GPE
Moscow → GPE
first → ORDINAL
Putin → PERSON
Ukraine → GPE
Just five weeks ago → DATE
Kazakhstan Kassym-Jomart Tokayev → ORG
Putin → PERSON
Putin → PERSON
Modi → GPE
Kremlin → ORG
Russia → GPE
Ukraine → GPE
Russia → GPE
more than four → CARDINAL
Kremlin → ORG
Ukraine → GPE
Russians → NORP
Russian → NORP
Reuters → ORG
Ukraine → GPE
Boryspil → GPE
Kyiv → PERSON
overnight → TIME
Russian → NORP
Four → CARDINAL
overnight → TIME
Russian → NORP
Ukraine → GPE
Boryspil → PERSON
recent days → DATE
Russia → GPE
Ukraine → GPE
Russia → GPE
Ukraine → GPE
Kommersant → PERSON
this morning → TIME
Russian → NORP
Kyiv → PERSON
Putin → PERSON
```

```
displacy.render(
    doc,
    style="ent",
    jupyter=True
)
```